Shaniya Jarrett

Curriculum Vitae

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Education

University of Maryland, College Park, MD — 2024 - Present
Doctor of Philosophy in Astronomy & Astrophysics

Fisk University, Nashville, TN — 2022 - 2024
Master of Science in Physics

University of Colorado at Boulder, Boulder, CO — 2014 - 2020
Bachelor of Arts in Astrophysics

Honors/Awards

- Finalist, Learner Tutoring Innovators of Color in STEM Scholarship — 2026
- Outstanding Teaching Assistant Award for Excellence in Teaching — 2025
- Margaret Cuninggim Women's Center Leadership Award — 2024
- Computer, Mathematical, and Natural Sciences (CMNS) Exceptional Recruitment Award — 2024
- Gregor and Donat Wentzel Scholarship — 2024
- Astronomy & Astrophysics Inclusive Excellence Award, UC San Diego (Declined) — 2024
- Chambliss Astronomy Achievement Student Award — 2023
- 24th Annual Fisk Research Symposium, Best Poster Presentation — 2023
- Si Se Puede Scholarship — 2023
- General Atomics Spotlight Award — 2022
- General Atomics Mission of Excellence Award — 2021, 2022
- University of Colorado at Boulder, Dean's List — 2016

Fellowships/Grants

- National Science Foundation Graduate Research Fellowship Program (GRFP) — 2024
- Graduate School Dean's Fellowship, University of Maryland, College Park — 2024
- Provost's Graduate Fellowship - Russell G. Hamilton Scholar, Vanderbilt University (Declined) — 2024
- Heising-Simons Foundation's Women & Girls in Astronomy Grant — 2023
- Inclusive Graduate Education Network (IGEN) Travel Grant — 2023
- Fisk – Vanderbilt Bridge Fellowship — 2022

Research Experience

NASA Goddard Space Flight Center, Greenbelt, MD **-** *Graduate Research Internship,* 06/2025 - Present

- Advisor: Scott Noble
  - Developing and fine-tuning a neural network classification pipeline to detect binary black hole signatures within AGN light curves.
  - Leveraging pre-trained audio transformers and HuggingFace tools to build an interpretable machine learning framework for astrophysical time-series data.

University of Maryland, College Park, MD
Graduate Researcher, 10/2024 - Present

- Advisor: Cole Miller
  - Exploring the use of multimodal machine learning models and advanced neural network architectures such as transformers and CNNs for applications in gravitational wave (GW) research.

Vanderbilt University, Nashville, TN
Graduate Researcher, 12/2022 - 2024

- Advisor(s): Kelly Holley-Bockelmann, Robert Scherrer
  - Created maps of the Gravitational Wave Background for the Laser Interferometer Space Antenna (LISA) mission.

- Used the two-point correlation function to determine if random data correlations can be distinguished from using galaxy distributions as a proxy for binary blackhole (BBH) mergers.

Center for Astrophysics and Space Astronomy (CASA), Boulder, CO
Laboratory Technician/Research Assistant, 01/2020 - 08/2020
- Advisor: Dmitry Vorobiev
  - Modeled digital micromirror devices (DMDs) for the SPRITE CubeSat.
  - Tested DMD theoretical performance on a multi-object spectrograph for beam size vs. projected angle.

Laboratory of Atmospheric and Space Physics (LASP), Boulder, CO
Research Assistant, 01/2019 – 08/2019
- Advisor(s): Xiaohua Fang, Yaxue Dong
  - Studied variability of Mars magnetic field interactions and compared MAVEN data to an empirical model.
  - Built projective global maps for Mars intrinsic field at various altitudes in orbit and performed statistical analysis.

Research Assistant, 01/2018 - 09/2018
- Advisor: Steven Cranmer
  - Identified solar wind irregularities using the OMNI database.
  - Developed algorithms to detect Coronal Mass Ejections.

Research Experience for Undergrads (REU) Internship, 05/2017 – 08/2017
- Advisor: David Malaspina
  - Analyzed data from the Magnetospheric Multiscale Mission to study solitary waves in Earth's magnetosphere and identify current density correlations.

Related Experience

General Atomics, Centennial, CO
Satellite Test Engineer, 08/2020 - 08/2022
- Performed functional testing and troubleshooting of systems, software, and hardware for the entire lifecycle of the $65 million Gazelle spacecraft.
- Integrated the Argos-4 instrument hosted by NOAA and France's National Center for Space Studies (CNES).
- Pioneered the company's first successful thermal vacuum test campaign.
- Trained and prepared new Test Engineers on spacecraft test functions.
- Led the test campaign for Vibration and Mass Properties testing.
- Presented technical data and final reviews to customers.
- Applied technical knowledge to develop test procedures and implement solutions for test data discrepancies.

Independent Data Analytics Project — Women-Led Podcast Analysis, 07/2026
- Analyzed public YouTube data from 325 episodes across eight women-led podcasts to examine how episode format, topic, duration, and titles relate to audience reach and interaction.
- Applied statistical testing, bootstrap confidence intervals, and within-channel normalization to distinguish reach from engagement per viewer.
- Developed a public interactive Plotly report and translated findings into practical content-performance recommendations.
- **Interactive report:** https://shaniyasolves.github.io/women-led-podcast-analytics/

Certifications

Microsoft, *Agent-a-thon: Build AI Agents in Copilot Studio*, 03/2026
CITI Program, *Responsible Conduct of Research* (RCR), 01/2023, 08/2025
General Atomics, *Electrostatic Discharge Control* (ESD), 01/2021

Outreach

- *Program Manager/Creator,* AstroBeats Project (Check out Astrobeats.us for music) 2023-2024
  - Guided a cohort of 8 middle school girls to create music using Astrophysical data.
  - Developed lecture materials for the 4-day workshop on various topics in Astronomy.
  - Planned and coordinated the resulting Dyer Observatory event where an ensemble performed their songs.
  - Received recognition from the College of Arts and Sciences Dean's Office at Vanderbilt and was covered in their Student Spotlight news story in February 2024.
- *Instructor,* Black Girls Becoming (BGB) Summer Institute 2023
  - Guided a cohort of 20 7^{th}-8^{th} grade girls through a transformative 2-week STEM program.
  - Developed lecture materials as the only graduate student instructor among Vanderbilt Professors.

- Championed the introduction of Astrophysics into the institute's curriculum, which will benefit future initiatives in our field as this program is a part of an IRB-approved nine-year study.
- *Poster Judge,* Research Experience for High School Students (REHfSS) – Summer Research Symposium through the School for Science and Math at Vanderbilt (SSMV) 2023
  - Evaluated research projects from 35+ students, many of whom are from marginalized backgrounds.
- *Guest Lecturer*, Vanderbilt Collaborative for STEM Education & Outreach (CSEO) 2023
  - Lectured on "Dark Matter & Orbital Dynamics" and created an interactive group project.
  - Led a three-hour session for 20 middle school students.
- *Volunteer Tutor*, School for Science & Math at Vanderbilt (SSMV) through CSEO 2022 - 2023
  - Offers free bi-weekly tutoring to underrepresented students in Physics.
- *Outreach Volunteer*, Vanderbilt Dyer Observatory 2022 - 2023
  - Offers knowledge on telescopes and space to foster interest in STEM for students and the general public.
  - Helped facilitate two telescope nights bringing in over 100 people/event.
- *Organization Member*, CU Boulder Science Technology & Astronomy Recruits (STARs) 2019 - 2020
  - Provide kids in rural areas with access to topics on Physics/Astronomy.
  - Created and led 4 six-hour lessons plans facilitating over 60 students/event.
- *Outreach Volunteer*, CU Boulder Fiske Observatory 2018 - 2019

Academic/Professional Service & Development

Academic:

University of Maryland, College Park

- *Teaching Assistant,* ASTR 100-Introduction to Astronomy with Prof. J. Hunt and Prof. R. Mushotzky 2024–2025
- *Workshop Attendee*, *Scientific Machine Learning for Gravitational Wave Astronomy*, ICERM 2025
  - Explored neural network–based machine learning techniques for gravitational wave detection, characterization, generation, and Bayesian approaches to parameter estimation through invited talks, hands-on tutorials, and networking with researchers in astrophysics and AI.
- *Workshop Guide,* Foundations of Astronomical Data Science 2025
  - Helped guide participants through learning Python, Bash, ADQL, SQL, and Gaia satellite data to build foundational data science skills for research.

Vanderbilt University

- *Program Member*, Establishing Inclusive Training for Multi-Messenger Astronomy (EMIT) 2022 - 2024
  - Training program for students dedicated to engaging in cross-disciplined research/coursework regarding Multi-Messenger Astronomy while building an inclusive community of scholars.
- *Executive Board Member*, Organization of Black & Professional Graduate Students (OBGAPS) 2022 - 2024
  - Facilitates the transition into post-baccalaureate programs, provides professional/ academic support, and creates networking opportunities.
  - Promotes concerns of our 60+ members to increase recruitment and retention of black students and faculty.

University of Colorado, Boulder

- *Co-Founder*, Women of Color in STEM (Astrophysics Branch) 2018 - 2020
  - Provides mentorship and resources for underrepresented minority women in STEM.
  - Coordinated the 1st Annual Women of Color in STEM Brunch which pulled 50+ students.
- *Research Committee Member,* Astronomy Club 2017 – 2019
  - Organized group research projects for 15+ students utilizing Fiske Observatory's telescopes and equipment.

Professional:

- *Executive Secretary*, National Aeronautics and Space Administration (NASA) Astrophysics Review Panel 2024
  - Summarized notes and comments throughout panel proceedings.
- *Attendee*, Legacy Survey of Space and Time (LSST) Data Workshop - Fisk University 2024
  - Led by Dr. Gloria Fonseca Alvarez from NOIRLab, the 3-day workshop provided participants with the skills to use the Rubin Science Platform.
  - Activities included interactive tutorials on Data Preview 0.2, catalog queries, and image display.
  - Collaborated on exercises for supernova light curves, host galaxies, and advanced computational techniques.

- *Selected Attendee,* Women of Color Project (W+OCP) – Harvard University 2023
  - One of 30 women selected out of 260 applicants to participate.
- *Planning Committee Member,* Fisk-Vanderbilt Bridge Program - Bridge Research Celebration Day (BRCD) 2023
  - Organized and coordinated the 11th Annual Research Celebration Day showcasing Bridge student's research achievements for 40+ attendees.
  - Worked to secure keynote speaker Adrian Parker, Missile Energy Systems Chief Engineer with the Naval Surface Warfare Center.
- *Selected Attendee,* Establishing Inclusive Training for Multi Messenger Astronomy (EMIT) Summer School in Multi-Messenger Astronomy 2023
  - Attendance based on students with demonstrated leadership potential in Multi-Messenger Astronomy.
  - Included sessions on MMA science, GW source & data analysis, electromagnetic counterparts, statistical techniques for MMA, topics in DEI, citizen science projects, and building community networks.
- *Selected Attendee,* National Society of Black Physicists - Student Leadership and Development Summit 2023, 2024
  - Attendance based on students with demonstrated leadership potential in Physics/Astronomy.
  - Summit designed to enhance Black student leadership acumen through interactive workshops on grantsmanship, project management, implicit bias, and mentorship.
  - Speakers included: Dr. William Wilson – Executive Director at the Harvard Center for Nanoscale Systems, Dr. Kent Wallace – Dean of Fisk University, Dr. Donnell Walton - Director of Corning Inc., Dr. Renee Horton – NASA Quality Engineer, Jesus Beltran - Mechanical Engineering Manager at Apple.
- *Organization Member,* General Atomics Black Employees (GABE) 2021 - 2022
  - Brings recognition and support to underrepresented groups.
- *Organization Member*, General Atomics Women's Internal Network (WiN) 2020 – 2022

Media/Public Engagement

- Featured in UMD News, *"Tuning into the Cosmos"* — March 2026
  - Highlighted research using transformer-based models to analyze gravitational wave signals and support future missions like LISA.
- Guest Speaker, *Pale Blue Dot Podcast* — January 2025
  - Discussed the AstroBeats Project and how music can be used to engage students with astrophysics concepts.
- Guest Speaker, *It Starts With A Bang Podcast* (Episode #117: "Gravitational Waves and the Universe") — May 2025
  - Explored the science of gravitational waves and how machine learning is transforming data analysis in astrophysics.

Publications

- *S. M. Jarrett*, "Comparing the Spatial Correlation of Binary Black Hole Mergers to Large-Scale Structure through the Illustris Simulation" 2025
- *Co-Author, "*Establishing the Infrastructure for a Collaborative Multi-messenger Ecosystem" White Paper 2023
- *Co-Author,* "Reimagining the Astronomy Ph.D. for the 21st Century" White Paper 2023

Presentations/Talks

- **Symposium:** S.M. Jarrett, "Detecting Binary Black Holes with Audio-Trained Transformers" presented at *NASA Goddard Space Flight Center,* August, 2025
- **Invited Speaker:** "The Future of Humanity in Space," "Education and Opportunity: Designing Sustainable and Inclusive Futures," and "Music and Astronomy," presented at the *Conference on World Affairs (CWA),* University of Colorado Boulder, April 2025
- **Conference**: S.M. Jarrett "Comparing the Spatial Correlation of Binary Black Hole Mergers to Large-Scale Structure through the Illustris Simulation," presented at the *245th American Astronomical Society (AAS),* January, 2025
- **Conference**: S.M. Jarrett "Harmonizing Astronomy and Music: Empowering Black Girls through the AstroBeats Project" presented at the *245th American Astronomical Society (AAS),* January, 2024, 2025
- **Keynote Speaker:** "Lessons and Insights for Young Researchers" presenting at the *School for Science and Math at Vanderbilt (SSMV) summer retreat*, June, 2024
- **Invited Speaker:** "Establishing Inclusive Training for Multi Messenger Astronomy," presenting at NOIRLab's *Windows on the Universe: Establishing the Infrastructure for a Collaborative Multi-messenger Ecosystem* workshop, October, 2023
- **Conference**: S.M. Jarrett "Leveraging Galaxy Distributions to Model the Astrophysical Foreground in Simulated LISA Data," presented at the *243rd American Astronomical Society (AAS),* January, 2023, 2024

- **Conference:** S.M. Jarrett "Unveiling Hidden Signals: Leveraging Galaxy Distributions to Model the Astrophysical Foreground in Simulated LISA Data," presented at *Regional Innovation Showcase with the National Science Foundation (NSF*), April, 2023
- **Symposium:** S.M. Jarrett "Determining the Spatial Distribution of Galaxies as a Proxy for Binary Black Hole Mergers in Simulated LISA Observations," presented at *24th Annual Fisk Research Symposium,* April, 2023
- **Conference:** S.M. Jarrett "Detecting Solitary Waves in Earth's Magnetosphere," presented at the *241st American Astronomical Society (AAS),* January, 2023
- **Showcase:** S.M. Jarrett, "Detecting Solitary Waves in Earth's Magnetosphere" presented at the *Laboratory for Atmospheric and Space Physics (LASP),* August, 2018

Professional Memberships/Affiliations

National Society of Black Engineers (NSBE), *Member*, 2026
Laser Interferometer Space Antenna (LISA) Consortium, *Member*, 2024-Present
National Society of Black Physicists (NSBP), *Member*, 2023-Present
African American Women in Physics (AAWIP), *Member*, 2023-Present
American Physical Society (APS), *Member*, 2023-Present
American Association for the Advancement of Science (AAAS), *Member*, 2023
American Astronomical Society (AAS), *Member*, 2022-Present

Skillset

Technical Skills:

- Python
- Atlassian Tools
- IDL programming
- Mathematica
- LaTeX
- SQL
- TensorFlow
- Deep Learning
- Git/GitHub
- Atlassian Tools
- SolidWorks
- Zemax
- Lumerical
- PuTTy (Terminal Emulator)
- LaTeX
- AWS
- APIs
- Tableau

Selected Coursework:

- Astrostatistics
- Computational Techniques
- Data Analysis
- Mathematical Methods
- Quantum Computing
- Computational Astrophysics
- Scientific Writing
- Communication & Conflict Management
- Advanced Quantum Mechanics
- Advanced Electrodynamics
- Advanced Mechanics
- Astronomical Instrumentation